

UPI Transaction Pattern Analysis — Churn Signals & Product Recommendations

Business Requirements Document

Project Title: UPI Transaction Pattern Analysis — Churn Signals & Product Recommendations

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SECTION 1 — BUSINESS BACKGROUND

UPI has grown from 2 crore transactions in FY 2016–17 to over 24,162 crore transactions in FY 2025–26 — a 12,000-fold increase in a decade. At NPCI's current scale of 698 million daily transactions, even a 1–2% failure rate translates to millions of failed transactions every single day. When users experience repeated failures on one app but succeed on a competitor's, they switch platforms and rarely return.

This analysis examines 250,000 synthetic UPI transactions spanning January to December 2024, modeled on real NPCI and RBI usage patterns. The dataset covers transaction types (P2P, P2M, Bill Payment, Recharge), merchant categories, sender banks, device types, network types, geographic states, and age groups.

Dataset is synthetically generated to simulate real UPI transaction patterns while preserving privacy compliance — inspired by NPCI and RBI market data. Source: Kaggle UPI Transactions 2024 Dataset. Macro reference: NPCI Monthly Transaction Statistics and India Data Portal.

SECTION 2 — BUSINESS OBJECTIVE

This analysis aims to:

1. Identify transaction failure patterns by type, category, hour, day, bank, device, and network to enable targeted infrastructure and UX improvements.
2. Understand P2P vs P2M adoption split and identify the merchant payment adoption gap.

3. Segment users by behavioral persona and age group to support personalized product interventions.
 4. Identify geographic and temporal failure hotspots for proactive engineering action.
 5. Deliver three specific product recommendations to reduce failure-driven churn and increase P2M adoption.
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SECTION 3 — STAKEHOLDERS

Stakeholder	Role	Interest
Product Manager	Primary	Feature prioritization based on failure findings
Growth Team	Primary	User re-engagement and dormancy prevention strategies
Engineering Team	Secondary	Peak hour and bank-level failure data for infrastructure planning
UX Team	Secondary	Failure message redesign based on timing and pattern data
Business Leadership	Approver	Overall platform health and churn risk view

SECTION 4 — SCOPE

In Scope:

Transaction failure analysis by type, category, hour, day, bank, device, and network. P2P vs P2M adoption analysis across volume and value. Age group behavioral analysis including ticket size and P2M adoption rate. Geographic failure rate analysis by sender state. Fraud flag pattern analysis covering 480 flagged transactions. Transaction value band distribution from Micro to Large. Three product recommendations with data-backed justification. Tableau dashboard with three views covering Platform Health, Time Patterns, and Segments.

Out of Scope:

Real-time transaction monitoring. Bank-level infrastructure root cause diagnosis. Fraud detection modeling or scoring. Individual user PII or account-level details. Revenue or profitability modeling.

Assumptions:

Dataset of 250,000 transactions spanning January to December 2024 is representative of platform behavior. Synthetic data is modeled on real NPCI and RBI usage patterns. Failure reasons are analyzed at transaction type and bank level, not individual API call level. All monetary values are in Indian Rupees (INR).

SECTION 5 — KEY FINDINGS

Platform Health

Metric	Value
Total Transactions Analyzed	2,50,000
Successful Transactions	2,37,624
Failed Transactions	12,376
Overall Failure Rate	4.95%
Total Value Processed	Rs.32.79 Crore
Average Ticket Size	Rs.1,311.76

Failure by Merchant Category

Rank	Category	Failure Rate	Avg Ticket Size
1	Education	5.25%	Rs.5,094
2	Shopping	5.09%	Rs.2,573
3	Food	5.01%	Rs.532
4	Grocery	5.01%	Rs.1,166
5	Other	4.95%	Rs.849
6	Entertainment	4.92%	Rs.413
7	Utilities	4.86%	Rs.2,361
8	Healthcare	4.83%	Rs.543
9	Fuel	4.80%	Rs.1,555
10	Transport	4.76%	Rs.308

Failure by Time

Time Pattern	Value
Peak Failure Hour	6 AM — 5.40% failure rate
Second Highest Failure Hour	Midnight — 5.37% failure rate
Peak Volume Hour	7 PM — 21,232 transactions
Worst Day of Week	Sunday — 5.10% failure rate
Best Day of Week	Friday — 4.77% failure rate
Worst Month	March 2024 — 5.38% failure rate
Best Month	October 2024 — 4.74% failure rate

Failure by Bank

Bank	Total Transactions	Failure Rate
Yes Bank	24,860	5.10%
ICICI	29,769	5.04%
Kotak	20,032	4.98%
IndusInd	25,173	4.95%
Axis	25,042	4.95%
SBI	62,693	4.94%
PNB	24,946	4.89%
HDFC	37,485	4.82%

Failure by Device and Network

Device + Network	Failure Rate
Web + 3G	6.60% — Worst combination
Android + 3G	5.23%
Web + 4G	5.11%
iOS + 4G	4.99%
Android + 4G	4.97%
iOS + WiFi	4.72% — Most reliable

Transaction Type Split

Type	Count	% of Volume	% of Value
P2P	1,12,445	44.98%	44.87%
P2M	87,660	35.06%	35.29%
Bill Payment	37,368	14.95%	14.91%
Recharge	12,527	5.01%	4.93%

Age Group Analysis

Age Group	Total Transactions	Avg Ticket	Failure Rate	P2M Adoption
18–25	62,345	Rs.1,194	4.89%	34.97%
26–35	87,432	Rs.1,326	4.84%	35.23%
36–45	62,873	Rs.1,424	5.13%	35.06%
46–55	24,841	Rs.1,333	5.06%	35.16%
56+	12,509	Rs.1,188	4.92%	34.27%

Geographic Analysis (Top 6 States)

State	Total Transactions	Failure Rate
Maharashtra	37,427	4.92%
Uttar Pradesh	30,125	5.22%
Karnataka	29,756	4.86%
Tamil Nadu	25,367	5.12%
Delhi	24,870	4.97%
Telangana	22,435	4.71%

Value Band Distribution

Band	Count	% of Total	Failure Rate
Micro (Under Rs.500)	1,06,462	42.58%	4.87%
Small (Rs.500–Rs.5,000)	1,32,582	53.03%	4.98%
Medium (Rs.5,001–Rs.25,000)	10,919	4.37%	5.31%
Large (Above Rs.25,000)	37	0.01%	13.51%

SECTION 6 — FUNCTIONAL REQUIREMENTS

FR#	Requirement	Priority
FR01	Calculate overall platform failure rate broken down by transaction type and merchant category	High
FR02	Display hourly and day-of-week failure patterns to identify peak degradation windows	High
FR03	Analyze failure rates by sender bank to identify bank-level infrastructure issues	High
FR04	Show P2P vs P2M split by volume and value to quantify the merchant adoption gap	High
FR05	Segment users by age group including average ticket size and P2M adoption rate	Medium

FR06	Identify geographic failure hotspots by sender state	Medium
FR07	Analyze device type and network type impact on failure rates	Medium
FR08	Show transaction value band distribution to understand platform usage patterns	Medium
FR09	Identify fraud transaction patterns from flagged records	Low

SECTION 7 — USER STORIES

US01 — Failure Visibility

As a Product Manager, I want to see which merchant categories have the highest failure rates so that I can prioritize engineering fixes that protect daily-use categories like Food and Grocery.

Acceptance Criteria: All merchant categories ranked by failure rate. Platform average reference line at 4.95%. Categories above average clearly identifiable in the dashboard.

US02 — Peak Hour Awareness

As an Engineering Lead, I want to see transaction volume and failure rate by hour of day so that I can schedule bank server health checks before known peak failure windows.

Acceptance Criteria: 24-hour view showing volume bars colored by failure rate. 6 AM and midnight failure spikes clearly identifiable. Day-of-week view showing weekend degradation pattern.

US03 — Bank Failure Intelligence

As a Product Manager, I want to see which sender banks have the highest failure rates so that I can build a Bank Health Indicator feature prioritizing the most affected banks.

Acceptance Criteria: All 8 banks ranked by failure rate. Yes Bank at 5.10% and HDFC at 4.82% contrast clearly visible. Platform average reference line shown for comparison.

US04 — P2M Adoption Gap

As a Business Development Manager, I want to see the P2P vs P2M transaction split so that I can understand how far users are from merchant payment behavior and design incentives accordingly.

Acceptance Criteria: P2P and P2M shown as percentage of total volume and total value. All four transaction types shown for full context. Gap between P2P and P2M share clearly quantified.

US05 — Segment Targeting

As a Growth Manager, I want to see users grouped by age segment with failure rates and average ticket sizes so that I can design targeted re-engagement campaigns for the highest-value segments.

Acceptance Criteria: Five age groups shown with transaction count and average ticket size. 36–45 segment highlighted as high-value but high-failure. P2M adoption rate visible per segment.

SECTION 8 — PRODUCT RECOMMENDATIONS

Recommendation 1 — Bank Health Indicator

Finding: Yes Bank users face 5.10% failure rate compared to HDFC's 4.82%. Web plus 3G combinations reach 6.60% failure — nearly 33% above platform average. Large transactions above Rs.25,000 fail at 13.51%, nearly 3x the platform average of 4.95%.

Recommendation: Build a Bank Health Indicator feature — display real-time bank server status on the payment screen before the user initiates a transaction. If their bank is degraded, show an alternative payment suggestion or a pre-emptive warning message. Flag 3G network users with a nudge to switch to WiFi for better reliability.

Expected Impact: Reduces failure-driven churn in high-failure bank segments. Converts a silent failure risk into a trust signal — users feel informed rather than abandoned. Android-first development priority given 75.11% Android user share.

Recommendation 2 — Contextual Failure Messaging

Finding: Failure spikes to 5.40% at 6 AM and 5.37% at midnight — both are bank maintenance windows. Weekends show 5.09–5.10% failure compared to weekday average of 4.87%. Users in Uttar Pradesh and Tamil Nadu face 5.12–5.22% failure rates. All users currently see a generic error with no context or recovery guidance.

Recommendation: Replace the generic Transaction Failed message with a contextual one: "Your bank is experiencing high traffic right now. Try again in 2 minutes or switch to an alternative payment method." Add a one-tap retry button with a 90-second countdown timer. Customize the message based on hour and day failure patterns identified in this analysis.

Expected Impact: Converts a frustrating failure into a trust signal. Users feel informed rather than abandoned, directly reducing uninstall rates after failure events.

Recommendation 3 — P2M Re-engagement for 36–45 Segment

Finding: The 36–45 age group has the highest average ticket size at Rs.1,424 but the worst failure rate at 5.13% on the platform. P2M adoption is only 35.06% platform-wide. Education and Shopping categories — where this age group spends the most — show above-average failure rates of 5.25% and 5.09% respectively.

Recommendation: Target 36–45 high-value users with a Rs.25 cashback on their first P2M transaction in the Education or Shopping category. Route the push notification directly into the P2M payment flow, not the home screen. Fix the failure rate for this segment first — users with money to spend will not return after repeated bad experiences.

Expected Impact: Users who complete one P2M transaction show significantly higher 30-day retention than pure P2P users. Fixing the 36–45 failure experience unlocks the platform's highest-value segment for merchant revenue growth.

SECTION 9 — DELIVERABLES

#	Deliverable	Description
1	SQL Analysis	14 queries in SSMS covering all analysis dimensions
2	Excel Workbook	Summary metrics, category analysis, monthly trend, bank analysis
3	Tableau Dashboard	3 tabs: Platform Health, Time Patterns, Segments and Recommendations
4	GitHub Repository	SQL files, Excel workbook, documentation, dashboard screenshots
5	BRD Document	This document

SECTION 10 — TOOLS USED

Tool	Purpose
SQL Server Management Studio (SSMS)	Data loading and all 14 SQL queries
Microsoft Excel	Pivot analysis, summary metrics, conditional formatting
Tableau Public	Interactive 3-tab dashboard with 12 individual chart sheets
GitHub	Version control and project portfolio hosting
